Skills Summary

Multi-Step Linear Equations with Rational Coefficients

Use this reference while completing your worksheet or when studying.

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1. Clear the fractions with the LCD

Rule. Multiplying both sides of an equation by the same nonzero number gives an *equivalent* equation — one with exactly the same solution. So pick the **LCD** of every denominator and multiply *each term on both sides* by it. Every fraction disappears at once.

Order of moves: clear fractions → distribute → combine like terms → undo $+$ and $-$ → undo $\times$ and $\div$.

Example: Solve $\dfrac{x}{2} + \dfrac{x}{5} = 7$.

Step 1. Two different denominators, so clear them both at once instead of combining fractions: the LCD of 2 and 5 is 10.

Step 2. Multiply every term by 10: $5x + 2x = 70$, so $7x = 70$.

$\Rightarrow$ $\boldsymbol{x = 10}$

Try it: Solve $\dfrac{x}{3} + \dfrac{x}{4} = 14$.

check: $\boldsymbol{x = 24}$

2. Variable on both sides

Rule. Clear the fractions first — multiply *all four* terms by the LCD, the ones without fractions too. Then move every variable term to one side and every constant to the other, and finish with one division.

A term such as $\dfrac{x+2}{3}$ has the whole numerator over 3, so multiplying by the LCD keeps it together: $6 \cdot \dfrac{x+2}{3} = 2(x+2)$.

Example: Solve $\dfrac{3}{5}x - 2 = \dfrac{x}{10} + 1$.

Step 1. Variables sit on both sides and both sides carry fractions, so clear with the LCD of 5 and 10, which is 10, before collecting anything.

Step 2. Multiply every term by 10: $6x - 20 = x + 10$. Subtract x, add 20: $5x = 30$.

$\Rightarrow$ $\boldsymbol{x = 6}$

Try it: Solve $\dfrac{5}{6}x - 1 = \dfrac{x}{3} + 4$.

check: $\boldsymbol{x = 10}$

3. Inequalities with rational coefficients

Rule. Solve an inequality exactly like an equation, with one extra rule: multiplying or dividing both sides by a **negative** number *reverses* the symbol ($>$ becomes $<$, $\geq$ becomes $\leq$). Adding or subtracting never reverses it, and multiplying by a positive number never does either.

Check: test one number from your answer and one number outside it.

Example: Solve $-\frac{3}{4}x + 2 \geq 8$.

Step 1. The coefficient of x is negative, so plan on turning the symbol at the final step — everything before that step is ordinary equation work.

Step 2. Subtract 2: $-\frac{3}{4}x \geq 6$. Multiply by $-\frac{4}{3}$ and *reverse* the symbol.

$\Rightarrow \quad \boldsymbol{x \leq -8}$

Try it: Solve $-\frac{2}{5}x - 1 > 3$.

check: $\boldsymbol{x < -10}$

(!) Watch out: when you multiply by the LCD, multiply *every* term, including the plain numbers — forgetting the constant is the most common slip. And reverse the inequality symbol only when the number you multiply or divide by is negative, never because the answer itself is negative.